

OpenGRF: free OpenSim API tool for Ground Reaction Forces prediction

Authors: Andrea Di Pietro and Francesca Di Puccio, Department of Civil and Industrial Engineering, University of Pisa, (IT)

Contacts: andrea.dipietro@phd.unipi.it

Instructions for Running OpenGRF in MATLAB:

1. Set the OpenSim API Environment Variables

Follow the instructions provided in the OpenSim documentation to set the necessary environment variables for MATLAB:

[Scripting with MATLAB - OpenSim Documentation](#)

After setting the environment variables, restart MATLAB to apply the changes.

2. Prepare MATLAB Folder

Launch your current version of MATLAB and create a new folder where you will store the following files:

- **OpenGRF_v1.p**
- **load_sto.p**
- **load_mot.p**

Ensure that all three files are placed in this folder for easy access.

3. Run the OpenGRF Script

Open the working folder in MATLAB, Right-click on the **OpenGRF_v1.p** file and select **Run** from the context menu in MATLAB. This will execute the OpenGRF script.

4. Select the Input Files

OpenGRF will prompt you to select the following files:

- **Scaled/Subject-specific MSK model (.osim):** This is the OpenSim model of the subject you wish to analyze.
- **Kinematics file (.mot):** Select the kinematics file in .mot format that contains the motion data you want to analyze (e.g., joint angles).

5. Configure the Analysis Parameters

In the OpenGRF GUI, adjust the following parameters:

- **Start and End Time:** Set the desired time range for the analysis.
- **Low-Pass Cut-off Frequency:** Choose the appropriate cut-off frequency for the low-pass filter to apply to the kinematics data.
- **Threshold of Force (N) to Reach Contact:** Set the force threshold (in Newtons) of contact elements. The default values are:
 - **400 N** for marker-based kinematics
 - **1000 N** for markerless kinematics

Important Note:

If you observe a loss of contact in your analysis (i.e., the contact forces drop to zero), you may need to **increase the threshold value**. This can help ensure that the contact is maintained in the analysis and prevent the algorithm from mistakenly interpreting a contact loss.

6. Output GRFs File

After the analysis is complete (contact calibration can take up to 10 min + time required for the GRF estimation), OpenGRF will generate a Ground Reaction Forces (GRFs) file in .mot format. This file will be saved in the same folder as your kinematics file.

Note: OpenGRF is currently a work-in-progress project. We encourage you to test and use it for research purposes, but kindly refrain from citing it until the associated paper is published